Dell PowerFlex 5.1.x/4.5.x and PowerFlex Manager 5.1.x

Software Upgrade Guide

June 2026
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

Copyright © Dell Inc. All Rights Reserved. Dell Technologies, Dell, and other trademarks are trademarks of Dell Inc. or its subsidiaries. Other trademarks may be trademarks of their respective owners.

Contents

Chapter 1: Introduction **5**
Common PowerFlex terms 6

Chapter 2: Revision history **7**

Chapter 3: Overview **8**

Chapter 4: Upgrading from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex Manager 5.1.x **9**
Upgrade checklist 9
Operating system requirements 10
Upgrade workflow from PowerFlex 5.0.x and PowerFlex Manager 5.0.x and above to PowerFlex 5.1.x and PowerFlex Manager 5.1.x 11
Upload the compatibility matrix file 12
Add a compliance file 12
Upgrading PowerFlex Manager 13
Upgrading PowerFlex Manager from a local repository path 13
Upgrade PowerFlex software 14

Chapter 5: Upgrading from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x **15**
Upgrade checklist 15
Operating system requirements 16
Upgrade workflow from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x 17
Upload the compatibility matrix file 18
Add a compliance file 18
Upgrading PowerFlex Manager 19
Upgrading PowerFlex Manager from a local repository path 19
Upgrade PowerFlex software 20

Chapter 6: Upgrading the storage data client on PowerFlex nodes **21**
Upgrade the storage data client on VMware ESXi based nodes 21
Prerequisites 21
Upgrade the storage data client during the VMware ESXi 8.x upgrade 21
Upgrade the storage data client during the VMware ESXi 8.x to 9.x upgrade 23
Upgrade storage data client on Windows-based server 28
Upgrade storage data client on Linux and AIX 28
Upgrade hardware 29

Chapter 7: Post-upgrade configuration tasks **30**
Configuring NVMe initiators as hosts connecting to PowerFlex system 30
Configure NVMe initiators on hosts for Linux-based systems 30
Configuring NVMe over TCP on a VMware ESXi compute-only node 34

Configuring system data networks, network sets, hosts, host groups, and enabling persistent discovery (Optional)..... 37
Configuring system data networks, network sets, hosts, host groups, and enabling persistent discovery (Optional)..... 39
Define system data networks..... 40
Define host group and assign hosts..... 40
Define network sets..... 40

1

Introduction

The *Dell PowerFlex 5.1.x/4.5.x and PowerFlex Manager 5.1.x Software Upgrade Guide* provides instructions to upgrade existing 4.5.x or 5.0.x PowerFlex deployments to PowerFlex 5.1.x.

Erasure coding is supported with PowerFlex core software 5.0.x and above. Mirroring is supported with PowerFlex core software 4.5.x.

Use this guide only for PowerFlex software updates. PowerFlex rack and PowerFlex appliance customers should not use this document for upgrades. For PowerFlex rack documentation, go to PowerFlex rack technical documentation. For PowerFlex appliance documentation, go to PowerFlex appliance documentation.

The target audience for this document includes Dell Technologies customers and services.

To upgrade an existing PowerFlex 5.0.x and PowerFlex Manager 5.0.x, go to the *Upgrading from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex Manager 5.1.x* section for detailed instructions on the following topics:

- Upgrading PowerFlex Manager to 5.1.x
- Upgrading PowerFlex storage
- Upgrading storage data clients

PowerFlex Manager 4.8.x includes the catalog core package 4.5.5 and above.

To upgrade an existing PowerFlex 4.5.x and PowerFlex Manager 4.8.x, go to the *Upgrading from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x* section for detailed instructions on the following topics:

- Upgrading PowerFlex Manager to 5.1.x
- Upgrading PowerFlex storage
- Upgrading storage data clients

To upgrade an existing PowerFlex 4.5.x and PowerFlex Manager 4.6.x system, go to the *Dell PowerFlex 4.5.x and PowerFlex Manager 4.8.x Software Install and Upgrade Guide* for detailed instructions on the following topics:

- Upgrading PowerFlex Manager to 4.8
- Upgrading PowerFlex storage
- Upgrading storage data clients

To upgrade an existing PowerFlex 3.6.x system, go to the *Dell PowerFlex 4.5.x and PowerFlex Manager 4.8.x Software Install and Upgrade Guide* for detailed instructions on the following topics:

- Deploying PowerFlex Manager
- Importing existing PowerFlex 3.6.x system
- Upgrading PowerFlex system
- Upgrading storage data clients

This release adds support for CloudLink 8.3.

CloudLink 8.3 and later versions are supported only on PowerFlex platforms.

The CloudLink version 8.3 or later software download files and documentation are in the following locations for PowerFlex software:

- PowerFlex Software Drivers & Downloads
- PowerFlex Software Manuals & Documents

NOTE:

CloudLink 8.2

The Key Management Interoperability Protocol (KMIP) server and machine encryption functionalities are available.

CloudLink 8.3

KMIP server and machine encryption functionalities remain operational but are deprecated.They are no longer covered under standard support and are planned to be disabled in a future release.

Dell Technologies strongly recommends planning a transition to supported alternative solutions to ensure continued functionality and support. For guidance on available options or migration planning, contact your Dell Technologies account representative.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Related information

Upgrading from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex Manager 5.1.x
Upgrading from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x

Common PowerFlex terms

The following table identifies the common PowerFlex terms and associated acronyms for PowerFlex 5.x using the erasure coding data protection method and PowerFlex 4.x using the mirroring data protection method.

Table 1. Common PowerFlex erasure coding and mirroring terms

Term	Acronym
device gateway target	DGWT
PowerFlex data server	PDS
management virtual machine	MVM
metadata manager	MDM
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT
management data store	MDS
storage data server	SDS
storage virtual machine	SVM
storage data replicator	SDR
system configuration reporter	SCR

The following table summarizes key PowerFlex mirroring terminology changes. References in the documentation will be updated over time.

Table 2. PowerFlex mirroring terminology changes

Term	Formerly known as
PowerFlex management cluster (ESXi)	PowerFlex management controller 2.0
PowerFlex management cluster (vSAN)	PowerFlex management controller 1.0
PowerFlex management cluster	PowerFlex management controller
Single ESXi PowerFlex management node	PowerFlex management node
PowerFlex management node	PowerFlex controller node

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

2

Revision history

Table 3. Revision history

Date	Document revision	Description of changes
June 2026	1.0	Initial release

3

Overview

This guide provides the supported upgrade paths for PowerFlex Manager, PowerFlex software, and software bundles with important considerations for operating system conversions.

The following table lists the PowerFlex versions and upgrade paths that are supported for PowerFlex 5.1.x/4.5.x and PowerFlex Manager 5.1.x:

Table 4. Upgrading to PowerFlex

Source version	Source catalog (RCM/IC)	Upgrade path	Target version PowerFlex Manager	Target version PowerFlex software	Target catalog	Storage architecture
PowerFlex 5.0.x and PowerFlex Manager 5.0.x	RCM: 3.9.0.0 IC: 50.390.00	Direct	5.1.x	5.1.x	RCM: 3.9.1.1 IC: 51.391.01	Erasure coding
PowerFlex 4.5.5.x and PowerFlex Manager 4.8.x	RCM: • 3.8.3.0 • 3.7.8.0 IC: • 48.383.00 • 48.378.00	Direct	5.1.x	4.5.5.2	RCM: 3.8.4.1 IC: 51.384.01	Mirroring
PowerFlex 4.5.4 or 4.5.3 or 4.5.2.1 and PowerFlex Manager 4.6.x	RCM: • 3.8.2.0 • 3.8.1.0 • 3.8.0.0 • 3.7.7.0 • 3.7.6.0 • 3.7.5.0 IC: • 46.382.00 • 46.381.00 • 46.380.00 • 46.376.00 • 46.375.00	PowerFlex Manager must initially be upgraded to 4.8.x, then 5.1.x Catalogs are expected to be direct upgrades.	5.1.x	4.5.5.2	RCM: 3.8.4.0 IC: 51.384.00	Mirroring

4

Upgrading from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex Manager 5.1.x

The following section explains how to upgrade from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex Manager 5.1.x.

Related information

Introduction

Upgrade checklist

The following table provides a checklist of items that are required to upgrade. Ensure that you have done everything on this list before proceeding with the upgrade. Then, perform the remaining preparation tasks in this guide.

Table 5. Upgrade checklist

Item	Checked
General	
All servers that are intended for hosting PowerFlex components and PowerFlex Manager meet system requirements. Ensure to validate target version resource (CPU, RAM, and disk space) requirements for virtual machines and increase accordingly if necessary.	-
Secure Boot on servers is not supported. Ensure that Secure Boot is disabled in the BIOS of each server.	-
All nodes must have unique hostnames.	-
To use secure authentication mode, ensure that the latest OpenSSL 64-bit v1.0.2 \ v1.1.1 is installed on all servers in the system.	-
Ensure that you know the following: ● IP addresses of primary and secondary MDMs ● Admin username and password for the primary MDM ● Operating system hostname and password of the primary MDM ● LIA password	-
For large-scale systems with up to 2048 storage data clients, ensure that Linux-based operating systems of servers that host MDM, device gateway target, and PowerFlex data server are configured to support large-scale systems (up to 2048 storage data clients).	-
A PowerFlex system must have all storage data clients in connected state. If the storage data client is not connected, the upgrade fails. To avoid a failure, remove the storage data client from the user interface.	-
The LDAP authentication method for LIA is not supported and is replaced by mTLS. Before upgrading, remove the LIA LDAP using the PowerFlex gateway.	-
Networking	
For a fully converged Linux-based system, follow the networking recommendations in the *Dell PowerFlex and PowerFlex Manager Technical Overview*.	-
Powerflex Manager and the installer node	

Table 5. Upgrade checklist (continued)

Item	Checked
The nodes which host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. SSH should be enabled on all the management nodes and should be on the same subnet.	-

NOTE: During a PowerFlex system upgrade, protected maintenance mode requires PowerFlex node size plus an additional 5% of the relevant storage pool size for a successful upgrade. Ensure that there is sufficient space available to avoid an upgrade failure.

Operating system requirements

The following table lists the PowerFlex operating system requirements.

For the most updated list of operating systems supported by PowerFlex, see the *Dell PowerFlex and PowerFlex Manager Support Matrix* on the Dell Technologies support site.

Table 6. Operating system requirements

Operating system	Requirement
Linux	Packages required for all components, all Linux flavors: • *numactl* • *libaio* • *wget* • *apr* • *libapr* • *python2-rpm or python3-rpm* • *yum-utils* Additional packages required for MDM components: • bash-completion (for SCLI completion) • Latest version of Python 2.x or Python 3.x (for newer operating system such as Red Hat Enterprise Linux 9.x or 10.x) • binutils • java-17-openjdk-headless Additional packages that PowerFlex data server requires: • smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils Additional packages that Diag-tools requires: • sar • sysstat For supported storage controllers, the following packages are also needed: • DELL: perccli • LSI raid controllers: storcli • HP raid controllers: hpssacli Additional packages required for PowerFlex data server components that contain SDPM/ NVDIMMs: • ndctl • jq • daxio

Table 6. Operating system requirements (continued)

Operating system	Requirement
	• libpmem To use the secure authentication mode, ensure OpenSSL 64-bit v1.1 is installed on all servers in the system. • `libopenssl1_1_x-x.x.x86_64.rpm` • `openssl1-1.x.x-x.x.x86_64.rpm` To use LDAP, ensure OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: • pciutils
Windows	• Only storage data client is a supported component on Windows. • To install storage data client, ensure that Security Update for Windows for x64-based Systems is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1 is not supported) • Visual C++ redistributable 2010 package, 64-bit
VMware	Compute-only (storage data client) support: VMware ESXi operating system: 8.0 and 9.0 (storage data client only)

Upgrade workflow from PowerFlex 5.0.x and PowerFlex Manager 5.0.x and above to PowerFlex 5.1.x and PowerFlex Manager 5.1.x

The following section explains how to upgrade from PowerFlex 5.0.x and PowerFlex Manager 5.0.x to PowerFlex 5.1.x and PowerFlex 5.1.x.

Upgrading includes the following activities:

1. Complete the upgrade checklist to prepare your environment for PowerFlex Manager 5.0.x.
2. Open the PowerFlex Manager user interface and from **Settings**, upload the 5.1.x.x software catalog bundle (under the **Upload Compliance File**), and compatibility management (GPG) file.
3. Upgrade PowerFlex Manager from the **Settings** page. For more information, see *Upgrading PowerFlex Manager* in related information.
4. Upgrade the **PowerFlex System** (resource name) from the **Resources** menu in PowerFlex Manager.
5. Complete the post-upgrade configuration tasks if required. For more information, see *Post-Upgrade configuration tasks* in related information.

NOTE: If the upgrade fails for any reason, rerun the upgrade operation to restart the upgrade process. The upgrade operation starts from the first node and takes a longer time to complete.

Related information

Upgrading PowerFlex Manager

Post-upgrade configuration tasks

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The compatibility matrix file (`cm-xxxxxxxx.gpg`) can be found in the `Software_Only_Complete_5.1.x_xxx.zip` file. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file, then click **Choose File** to select the GPG file.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from Dell Technologies connectivity services (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
   - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from Dell Technologies connectivity services (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services , you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `PowerFlex _Software_5.1.x_xxx.zip`

NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: `\\host\share\filename.zip`
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file.
   - Mark the target upgrade catalog as the default version for compliance checking.
5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for a VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upgrading PowerFlex Manager

This section explains how to upgrade PowerFlex Manager 5.0.x to a newer version.

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Related information

Upgrade workflow from PowerFlex 5.0.x and PowerFlex Manager 5.0.x and above to PowerFlex 5.1.x and PowerFlex Manager 5.1.x

Upgrading PowerFlex Manager from a local repository path

Upgrade the management software from a .TGZ file on a local repository path.

Prerequisites

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You can still force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

NOTE: PowerFlex Manager is unavailable during the upgrade. An API or any API calls attempted during the PowerFlex Manager upgrade are unsuccessful until the API gateway is available and operational.

Steps

1. Download the PowerFlex management platform .TGZ file.
2. Copy the PowerFlex management platform .TGZ file to a share drive location.
3. On the menu bar, click **Settings**, and then click **Software Upgrade**.
4. Click **Management Software Upgrade**.

   NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter (SCR) to validate that configurations are ready to upgrade. PowerFlex SCR is a standalone application designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

5. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
6. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
7. Click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
8. Under **Management Credentials**, optionally override the default username and password for the nodes hosting PowerFlex Manager. The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. You can enter the username for the first row in the table, and the changes you make will be applied to all the nodes. Enter the password for each of the nodes.
9. Enter **`UPDATE POWERFLEX MANAGER`** if you want to proceed with the update. If you want to perform an update even if none is required, enter **`FORCE UPDATE`**.
10. Click **Yes** to update PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to log in once again, after the upgrade process is complete.

Next steps

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Upgrade PowerFlex software

Use this procedure to upgrade PowerFlex software.

Steps

1. Log in to PowerFlex Manager.
2. To upgrade the PowerFlex gateway, on the **Resources** page, select the resource that has the **OS Hostname** as **powerflex**.
3. Verify the PowerFlex node details.
4. Click **Update Resources**.
5. On the **Summary** page, select **Allow PowerFlex Manager to perform non-disruptive updates now** and **Instant Maintenance Mode**. For more information about instant maintenance mode and protected maintenance mode, see the *Dell PowerFlex and PowerFlex Manager Technical Overview*.
6. Enter **`UPDATE POWERFLEX`** in the required field.
7. Click **Finish**.
8. Click **Yes** to confirm the changes.

5

Upgrading from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x

The following section explains how to upgrade from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x.

Related information

Introduction

Upgrade checklist

The following table provides a checklist of items that are required to upgrade. Ensure that you have done everything on this list before proceeding with the upgrade. Then, perform the remaining preparation tasks in this guide.

Table 7. Upgrade checklist

Item	Checked
General	
All servers that are intended for hosting PowerFlex components and PowerFlex Manager meet system requirements. Ensure to validate target version resource (CPU, RAM, and disk space) requirements for virtual machines and increase accordingly if necessary.	-
Secure Boot on servers is not supported. Ensure that Secure Boot is disabled in the BIOS of each server.	-
All nodes must have unique hostnames.	-
To use secure authentication mode, ensure that the latest OpenSSL 64-bit v1.0.2 \ v1.1.1 is installed on all servers in the system.	-
Ensure that you know the following: ● IP addresses of primary and secondary MDMs ● Admin username and password for the primary MDM ● Operating system hostname and password of the primary MDM ● LIA password	-
For large-scale systems with up to 2048 storage data clients, ensure that Linux-based operating systems of servers that host MDM, SDS, and SDR are configured to support large-scale systems (up to 2048 storage data clients).	-
A PowerFlex system must have all storage data clients in connected state. If the storage data client is not connected, the upgrade fails. To avoid a failure, remove the storage data client from the user interface.	-
The LDAP authentication method for LIA is not supported and is replaced by mTLS. Before upgrading, remove the LIA LDAP using the PowerFlex gateway.	-
Networking	
For a fully converged Linux-based system, follow the networking recommendations in the *Dell PowerFlex and PowerFlex ManagerTechnical Overview*.	-
Powerflex Manager and the installer node	

Table 7. Upgrade checklist (continued)

Item	Checked
The nodes which host PowerFlex Manager must have connectivity to both data and management PowerFlex networks. SSH should be enabled on all the management nodes and should be on the same subnet.	-

NOTE: During a PowerFlex system upgrade, protected maintenance mode requires PowerFlex node size plus an additional 5% of the relevant storage pool size for a successful upgrade. Ensure there is sufficient space available to avoid an upgrade failure.

Operating system requirements

The following table lists the PowerFlex operating system requirements.

For the most updated list of operating systems supported by PowerFlex, see the *Dell PowerFlex and PowerFlex Manager Support Matrix* on the Dell Technologies support site.

Table 8. Operating system requirements

Operating system	Requirement
Linux	Packages required for all components, all Linux flavors: • *numactl* • *libaio* • *wget* • *apr* • *libapr* • *python2-rpm or python3-rpm* • *yum-utils* Additional packages required for MDM components: • bash-completion (for SCLI completion) • Latest version of Python 2.x or Python 3.x (for newer operating system such as Red Hat Enterprise Linux 9.x) • binutils • java-17-openjdk-headless Additional packages that storage data server requires: • smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils Additional packages that Diag-tools requires: • sar • sysstat For supported storage controllers, the following packages are also needed: • DELL: perccli • LSI raid controllers: storcli • HP raid controllers: hpssacli Additional packages required for storage data server components that contain SDPM/ NVDIMMs: • ndctl • jq • daxio

Table 8. Operating system requirements (continued)

Operating system	Requirement
	• libpmem To use the secure authentication mode, ensure OpenSSL 64-bit v1.1 is installed on all servers in the system. • `libopenssl1_1_x-x.x.x86_64.rpm` • `openssl1-1.x.x-x.x.x86_64.rpm` To use LDAP, ensure OpenLDAP 2.4 is installed on all servers. Additional packages needed for LIA: • pciutils
Windows	• Only storage data client is a supported component on Windows. • To install storage data client, ensure that Security Update for Windows for x64-based Systems is installed. To use the secure authentication mode, ensure that these are installed on all servers in the system: • OpenSSL 64-bit v1.0.1 or later (v1.1 is not supported) • Visual C++ redistributable 2010 package, 64-bit
VMware	Compute-only (storage data client) support: VMware ESXi operating system: 8.0 and 9.0 (storage data client only)

Upgrade workflow from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x

The following section explains how to upgrade from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x.

Upgrading includes the following activities:

1. Complete the upgrade checklist to prepare your environment for PowerFlex Manager.
2. Open the PowerFlex Manager user interface and from **Settings**, upload the 5.1.x.x software catalog bundle (under the **Upload Compliance File**), and compatibility management (GPG) file.
3. Upgrade PowerFlex Manager from the **Settings** page. For more information, see the *Upgrading PowerFlex Manager* topic in related information.
4. Upgrade the **PowerFlex System** (resource name) from the **Resources** menu in PowerFlex Manager.
5. Complete the post-upgrade configuration tasks if required. For more information, see the *Post-Upgrade configuration tasks* topic in related information.

NOTE: If the upgrade fails for any reason, rerun the upgrade operation to restart the upgrade process. The upgrade operation starts from the first node and takes a longer time to complete.

Related information

Upgrading PowerFlex Manager

Post-upgrade configuration tasks

Upload the compatibility matrix file

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if the software catalog versions that are currently loaded are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The compatibility matrix file (`cm-xxxxxxxx.gpg`) can be found in the `Software_Only_Complete_5.1.x_xxx.zip` file. The file has a GPG extension and an associated compatibility version number.

Prerequisites

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services has been properly configured.

Steps

1. On the menu bar, click **Settings** and click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file, then click **Choose File** to select the GPG file.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

Steps

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from Dell Technologies connectivity services (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
   - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from Dell Technologies connectivity services (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services , you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `PowerFlex _Software_5.1.x_xxx.zip`

NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: `\\host\share\filename.zip`
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file.
   - Mark the target upgrade catalog as the default version for compliance checking.
5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for a VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Upgrading PowerFlex Manager

This section explains how to upgrade PowerFlex Manager 4.8.x to a newer version.

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Related information

Upgrade workflow from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x

Upgrading PowerFlex Manager from a local repository path

Upgrade the management software from a .TGZ file on a local repository path.

Prerequisites

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

About this task

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You can still force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

NOTE: PowerFlex Manager is unavailable during the upgrade. An API or any API calls attempted during the PowerFlex Manager upgrade are unsuccessful until the API gateway is available and operational.

Steps

1. Download the PowerFlex management platform .TGZ file.
2. Copy the PowerFlex management platform .TGZ file to a share drive location.
3. On the menu bar, click **Settings**, and then click **Software Upgrade**.
4. Click **Management Software Upgrade**.

   NOTE: Dell Technologies recommends using the PowerFlex System Configuration Reporter (SCR) to validate that configurations are ready to upgrade. PowerFlex SCR is a standalone application designed to provide comprehensive configuration reporting, health assessment, configuration checks, readiness/upgrade assessments checks, and log collection for PowerFlex systems.

5. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
6. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
7. Click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
8. Under **Management Credentials**, optionally override the default username and password for the nodes hosting PowerFlex Manager. The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. You can enter the username for the first row in the table, and the changes you make will be applied to all the nodes. Enter the password for each of the nodes.
9. Enter **`UPDATE POWERFLEX MANAGER`** if you want to proceed with the update. If you want to perform an update even if none is required, enter **`FORCE UPDATE`**.
10. Click **Yes** to update PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to log in once again, after the upgrade process is complete.

Next steps

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Upgrade PowerFlex software

Use this procedure to upgrade PowerFlex software.

Steps

1. Log in to PowerFlex Manager.
2. To upgrade the PowerFlex gateway, on the **Resources** page, select the resource that has the **OS Hostname** as **powerflex**.
3. Verify the PowerFlex node details.
4. Click **Update Resources**.
5. On the **Summary** page, select **Allow PowerFlex Manager to perform non-disruptive updates now** and **Instant Maintenance Mode**. For more information about instant maintenance mode and protected maintenance mode, see the *Dell PowerFlex and PowerFlex Manager Technical Overview*.
6. Enter **`UPDATE POWERFLEX`** in the required field.
7. Click **Finish**.
8. Click **Yes** to confirm the changes.

6

Upgrading the storage data client on PowerFlex nodes

The following procedures perform the following upgrades:

- Storage data client to different operating systems
- VMware system patches
- VMware ESXi 8.x to 9.x

Upgrade the storage data client on VMware ESXi based nodes

The following section explains the prerequisites and procedures for upgrading the storage data client (SDC) on VMware ESXi 8.x or VMware ESXi 9.x nodes.

Prerequisites

Before upgrading the VMware ESXi system ensure that your system meets the following:

- The system is fully operational and includes an MDM cluster.
- All SDSs, storage nodes, and SDCs are connected.
- The VMware vCenter version matches the target VMware ESXi version for this upgrade. For example, if you are upgrading the compute-only nodes to VMware ESXi 9.0, vCenter must be at version 9.0.
- If a VMware ESXi cluster is defined, DRS is configured and the vMotion network is operational on all PowerFlex nodes.

Perform all operations in the PowerFlex Manager as an admin user.

Upgrade the storage data client during the VMware ESXi 8.x upgrade

Perform the following steps on a single PowerFlex node to prepare it for the VMware ESXi upgrade. Once the node is prepared, proceed with the upgrade and restore the PowerFlex node to its operational state. After the PowerFlex node is fully operational, repeat the following procedure on the next PowerFlex node in the upgrade sequence.

About this task

VMware ESXi 8.x is supported by PowerFlex software 5.x/4.5.x and PowerFlex Manager 5.1.x.

Steps

1. Verify that the system is in an optimal state.
   a. In the PowerFlex Manager left pane, click **Monitoring** > **Alerts**, and then in the right pane confirm that no SDS disconnect messages appear.
   b. If the node is an MDM cluster member, in the left pane select **Resources**, click **PowerFlex System**, and in the right pane, click **View Details** that the state of the cluster is Normal.
   c. In the left pane, click **Nodes** > **Protection Domain** and then in the right pane verify that the storage data clients are connected.
2. For each storage data client, record the storage data client's IP address.
3. Log in to vCenter using the vSphere client, and complete the following actions:

a. Correlate the storage data client data IP address that you recorded in step 2 to the VMware ESXi host's relevant VMkernel IP address.

4. In vCenter, right-click the compute-only node and select **Maintenance Mode** > **Enter Maintenance Mode**.
5. In the **Confirm Maintenance Mode** dialog box, clear the **Move powered-off and suspended virtual machines to other hosts in the cluster** checkbox, and click **OK**.
   - If DRS and vMotion are configured, the VMs should start to migrate to other nodes.
   - If DRS is not configured, but vMotion is, migrate the VMs to other nodes manually and repeat the previous step to enter maintenance mode.
6. If a warning message appears requesting you to migrate the VMs, click **OK** and go to the next step. In this case, click **OK** and proceed with the procedure.
7. Using SSH, connect to the VMware ESXi node using the management IP address, and log in as root user with the relevant password.

Uploading the VMware ESXi patch depot and storage data client installation files to PowerFlex nodes

Perform the following procedure to transfer the VMware ESXi patch depot and storage data client installation files to PowerFlex nodes. Ensure that the storage data client installation files are available locally on each node or accessible through a shared datastore before initiating the VMware ESXi upgrade and storage data client installation.

Prerequisites

Ensure that you have a shared volume or datastore with at least 400 MB of available space to store the VMware ESXi upgrade package and storage data client package.

Steps

1. Download the relevant VMware upgrade package from Broadcom Support Portal and if required the storage data client package from the Dell Technologies support site.
2. Upload the downloaded files to the PowerFlex nodes.

   The following are example files:
   - VMware path depot: `VMware-ESXi-8.0.0.0-xxxxxxxx-depot.zip`
   - Storage data client 5.x package: `scaleio-sdc-esx8x_5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip`
   - Storage data client 4.5.x package: `scaleio-sdc-esx8x_4.5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip`

   NOTE: File names include the version and build number of the target release.

Upgrade VMware ESXi 8.x on the PowerFlex node

Perform the following procedure to upgrade the VMware ESXi 8.x operating system on the PowerFlex node.

Steps

1. To optionally list the profile of the VMware ESXi .zip archive, run the following:

```
esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/<VMWare.zip>
```

   For example:

```
esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/VMware-
ESXi-8.OU3d-24585383-depot.zip
```

   Output similar to the following displays:

```
Name Vendor Acceptance Level Creation Time Modification Time
---------------------------- ------------ ---------------- ------------------
ESXi-8.OU3d-24585383-no-tools VMware, Inc. PartnerSupported 2025-03-04T00:00:00
2025-03-04T00:00:00
```

```
ESXi-8.OU3d-24585383-standard VMware, Inc. PartnerSupported 2025-03-04T00:00:00
2025-03-04T00:00:00
```

2. Upgrade the VMware ESXi version, run the following command as a single line:

```
esxcli software profile update -p ESXi-8.OU3d-24585383-standard -d /vmfs/volumes/
sio_datastore/VMware-ESXi-8.OU3d-24585383-depot.zip
```

3. After the upgrade has completed successfully, the following output appears, followed by a list of the upgraded packages:
   Update result message:

```
The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Reboot Required: true
```

4. Run *reboot* to reboot the system.

Upgrade storage data clients on VMware ESXi 8.x (optional)

Perform the following procedure to upgrade the storage data client on VMware ESXi 8.x.

Prerequisites

Ensure that you have the required installation software package for your operating system of your storage data client, download the zipped software packages from the Dell Technologies support site.

NOTE: The following procedure is only applicable for unmanaged storage data clients that might require manual updates.

Steps

1. Copy the `.VIB` file to VMware ESXi.
2. Run the following command:

```
esxcli software vib update -d <path to ZIP>
```

   Where `<path to ZIP>` is the path to the storage data client ZIP file.

3. Reboot the server.

Post-upgrade steps

After you have upgraded the system to the wanted VMware ESXi and storage data client version, verify that you have upgraded all nodes to the wanted VMware ESXi and storage data client version.

Verify all datastore are mounted. After a VMware ESXi upgrade, some remain unmounted.

NOTE: This applies to any VMware ESXi upgrade, from 8.x to 9.x, 9.0 to 9.0.x, or other variations of 9.x to 9.x upgrade.

Upgrade the storage data client during the VMware ESXi 8.x to 9.x upgrade

Perform the following steps on a single PowerFlex node to prepare it for the VMware ESXi upgrade. The following procedure is only supported for PowerFlex 5.x/4.5.x and PowerFlex Manager 5.1.x.

Steps

1. Back up the storage data client GUID of the node to be upgraded.

a. View the complete list of storage data client settings for the scini driver:

```
esxcli system module parameters list -m scini | grep Ioctl
```

Output similar to the following displays:

```
IoctlIniGuidStr string B6681618-6BED-48C6-A90E-6D5F997DC025 Ini Guid, for example:
12345678-90AB-CDEF-1234-567890ABCDEF
IoctlMdmIPStr string 192.168.141.246,192.168.142.246 Mdms IPs, IPs for MDM in same
cluster should be comma separated.
```

NOTE: The GUID is displayed in the command output. Once the storage data client installation and the VMware ESXi upgrade are complete, this output information is required to validate the GUID and MDM virtual IP addresses.

b. Copy the entire command output and the storage data client IP address into Notepad, to identify the storage data client associated with the GUID.

For example:

```
10.118.221.156 SDC settings:
"IoctlIniGuidStr= B6681618-6BED-48C6-A90E-6D5F997DC025
IoctlMdmIPStr=192.168.141.246,192.168.142.246”
```

2. Put the VMware ESXi node into maintenance mode. Run the following command:

```
esxcli system maintenanceMode set--enable true
```

3. Remove the storage data client from the VMware ESXi 8.x node.

a. Using SSH, check the VMware ESXi version that is installed on the host:

```
esxcli system version get
```

Output similar to the following displays:

```
Product: VMware ESXi
Version: 8.x.x
Build: Releasebuild-xxxxxxx
```

b. Check the version of the storage data client:

```
esxcli software vib list |grep scaleio
```

Output similar to the following displays:

```
scaleio-sdc-esx8.x  5.0-0.864 EMC PartnerSupported  2026-03-23 host
```

c. Remove the storage data client from the VMware ESXi 8.x host using one of the following commands:

```
esxcli software vib remove --vibname scaleio-sdc-esx8.x
```

The output should display the results of the command. For example:

```
Removal Result
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Reboot Required: true
VIBs Installed:
VIBs Removed: VMware_bootbank_scaleio-sdc-esx8.x
VIBs Skipped:
```

d. Do NOT reboot.

Related information

Install a storage data client on a VMware Cloud Foundation or ESXi 9.x node

Upgrade VMware ESXi using vSphere lifecycle manager

Perform the following procedure to upgrade VMware ESXi using the lifecycle manager of the hosts in your environment.

About this task

The vSphere Lifecycle Manager (vLCM) baselines, previously called vSphere Update Manager (VUM), is being deprecated. You can manage the lifecycle of the hosts in your environment by using the vSphere Lifecycle Manager images.

NOTE: When upgrading a VMware Cloud Foundation (VCF) implementation, Dell Technologies recommends coordinating with Broadcom Support to determine the appropriate time to upgrade the storage data client.

Prerequisites

Before updating the VMware ESXi host ensure that you have upgraded the vCenter Server to version 9.x.

Steps

1. Log in to the vCenter Server.
2. Go to the vSphere Cluster that contains the VMware ESXi host you want to upgrade.
3. Click the **Updates** tab, then click **Image** > **Setup Image manually**.
   On the image configuration screen, select **ESXi 9.x** from the drop-down list.
4. Once the image setup is ready, click **Save**.
5. Click **Finish** when the image setup process completes.
6. Click **Check the Compliance** button to display hosts that are not compliant.
7. Click the **Remediate All** button to bring non-compliant hosts into compliance.
8. After some time, the VMware ESXi host restarts and loads the upgraded version of VMware ESXi.

Uploading the VMware ESXi patch depot and storage data client installation files to PowerFlex nodes

Perform the following procedure to transfer the VMware ESXi patch depot and storage data client installation files to PowerFlex nodes. Ensure that the storage data client installation files are available locally on each node or accessible through a shared datastore before initiating the VMware ESXi upgrade and storage data client installation.

Prerequisites

Ensure that you have a shared volume or datastore with at least 400 MB of available space to store the VMware ESXi upgrade package and storage data client package.

Steps

1. Download the relevant VMware upgrade package from Broadcom Support Portal and if required the storage data client package from the Dell Technologies support site.
2. Upload the downloaded files to the PowerFlex nodes.
   The following are example files:
   - VMware path depot: `VMware-ESXi-9.0.0.0-xxxxxxxx-depot.zip`
   - Storage data client packages:
     - `scaleio-sdc-esx9x_5.x.x.xxx-xOEM.xxx.x.xxxxxxxx.zip`
     - `scaleio-daemon-esx_5.x.x.xxx-x.zip`

   NOTE: File names include the version and build number of the target release. Beginning with VMware ESXi 9.x, the storage data client is provided in two separate packages.

Upgrade VMware Cloud Foundation or VMware ESXi to version 9.x on PowerFlex nodes

Perform the following procedure to upgrade VMware ESXi to version 9.x on PowerFlex nodes.

Steps

1. To optionally list the profile of the VMware ESXi .zip archive, run the following:

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/<VMWare.zip>
   ```

   For example:

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/VMware-
   ESXi-9.x-xxxxxxxx-depot.zip
   ```

   Output similar to the following displays:

   ```
   Name Vendor Acceptance Level Creation Time Modification Time
   ---------------------------- ------------ ---------------- ------------------
   ESXi-9.x-xxxxxxxx-no-tools VMware, Inc. PartnerSupported 2025-03-04T00:00:00
   2025-03-04T00:00:00
   ESXi-9.x-xxxxxxxx-standard VMware, Inc. PartnerSupported 2025-03-04T00:00:00
   2025-03-04T00:00:00
   ```

2. Upgrade the VMware ESXi version, run the following command as a single line:

   ```
   esxcli software profile update -p ESXi-9.x-xxxxxxxx-standard -d /vmfs/volumes/
   sio_datastore/VMware-ESXi-9.x-xxxxxxxx-depot.zip
   ```

3. After the upgrade has completed successfully, the following output appears, followed by a list of the upgraded packages:
   Update result message:

   ```
   The update completed successfully, but the system needs to be rebooted for
   the changes to be effective.
   Reboot Required: true
   ```

4. Run the following command to reboot the PowerFlex node:

   ```
   #reboot
   ```

Install a storage data client on a VMware Cloud Foundation or ESXi 9.x node

Perform the following procedure to install the storage data client on a VMware Cloud Foundation (VCF) 9 or ESXi 9.x node.

About this task

Install the storage data client with the appropriate parameters to connect them to an existing PowerFlex system. The following procedure is relevant for adding more storage data clients to an existing system and for adding storage data clients to a two-layer system during initial deployment activities.

Steps

1. On the VMware ESXi-based node where you are installing the storage data client, set the acceptance level. Run the following command:

   ```
   esxcli software acceptance set --level PartnerSupported
   ```

An example of the set acceptance level command is the following:

```
[root@localhost: /vmfs/volumes/695f99a6-92442b9d-d02c-9c63c097c914] esxcli software
acceptance set --level Partnersupported
Host acceptance level changed to 'Partnersupported'.
[root@localhost:/vmfs/volumes/695f99a6-92442b9d-d02c-9c63c097c914]
```

2. Install the storage data client agent package. Run the following command:

```
#esxcli software component apply -d /<PATH>/scaleio-daemon-esx_5.x.x.xxx-x.zip
```

The following is an example of the install storage data client agent package command:

```
#esxcli software component apply -d /vmfs/volumes/datastore1/scaleio-daemon-
esx_5.1.0.234-0.zip
```

```
[root@localhost: /vrlls/volumes/69Sf99a6-92442b9d-d02c-9c63c097c914] esxcli software
component apply -d /vrlls/volumes/datastorel/scaleio-daemon-esx_5.1.O.234-0.zip
Installation Result
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Components Installed: scaleio-daemon-esx_5.1.0.234-0
components Removed:
components Skipped:
Reboot Required: true
DPU Results:
[root@localhost:/vmfs/volumes/ 695!99a6-92442b9d-d02c-9c63c097c914]
```

Where <PATH> is the directory containing the storage data client agent package (.zip file).

3. Do NOT reboot.
4. Install the storage data client driver package. Run the following command:

```
#esxcli software component apply -d /<PATH>/scaleio-sdc-esx9x_5.x.x.xxx-
xOEM.xxx.x.xxxxxxxx.zip
```

The following is an example of the install storage data client driver package command:

```
#esxcli software component apply -d /vmfs/volumes/datastore1/scaleio-sdc-
esx9x_5.1.0.293-1OEM.900.0.24755229.zip
```

```
[root@local host: /vmfs/volumes/695f99a6-92442b9d-d02c-9c63c097c914] esxcli
software component apply -d /vmfs/volumes/datastore1/scaleio-sdc-esx9x_5.1.0.293-
lOEM.900.0.24755229.zip
Installation Result
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
components Installed: scaleio-sdc-esx9x_5.10.293-1OEM.900.0.24755229
Components Removed:
Components Skipped:
Reboot Required: true
DPU Results:
[root@localhost: /vmfs/volumes/695f99a6-92442b9d-d02c-9c63c097c914}
```

5. Obtain approval to reboot the VMware ESXi node, then perform a graceful reboot to load the storage data client driver. Run the following command:

```
#reboot
```

6. After the node comes back online, ensure that the scini module is loaded and that the GUID and MDM virtual IP addresses match the values that were collected before the upgrade, by running the following command:

```
esxcli system module parameters list -m scini|grep Ioctl
```

a. If the output from step 6 does not include the GUID and MDM virtual IP addresses, use the following example command, and verify that the values are replaced with your values. Restore the GUID and MDM virtual IP addresses using the details that were collected in an earlier procedure; otherwise, continue to step 7. See related information for more details.

```
esxcli system module parameters set -m scini -p
 "IoctlIniGuidStr=0d0cc350-08a3-49f9-9f02-54b17f416ffa
IoctlMdmIPStr=10.10.150.20,10.10.150.21,10.10.150.22"
```

Reboot the host again.

7. Exit the VMware ESXi host from maintenance mode.

```
#esxcli system maintenanceMode set--enable false
```

8. Perform the following script to persist configuration changes made at the command line.

```
#/bin/auto-backup.sh
```

Related information

Upgrade the storage data client during the VMware ESXi 8.x to 9.x upgrade

Post-upgrade steps

After you have upgraded the system to the wanted VMware ESXi and storage data client version, verify that you have upgraded all nodes to the wanted VMware ESXi and storage data client version.

Verify all datastore are mounted. After a VMware ESXi upgrade, some remain unmounted.

NOTE: This applies to any VMware ESXi upgrade, from 8.x to 9.x, 9.0 to 9.0.x, or other variations of 9.x to 9.x upgrade.

Upgrade storage data client on Windows-based server

Perform the following procedure to upgrade storage data client on a Windows-based server.

Prerequisites

Ensure that you have the required installation software package for your operating system of your storage data client, download the zipped software packages from the Dell Technologies support site.

Steps

1. Double-click the MSI file and click **Next** to finish.
2. Reboot the node.

Upgrade storage data client on Linux and AIX

Perform the following procedure to upgrade storage data client on Linux and AIX.

Prerequisites

Ensure that you have the required installation software package for your operating system of your storage data client, download the zipped software packages from the Dell Technologies support site.

NOTE: Do not reboot cluster members that are serving as part of a co-resident cluster after installing a new storage data client. For more information, see the *Dell PowerFlex and PowerFlex Manager Software Administration Guide*.

Steps

Install the appropriate package for your Linux distribution.

For example:

```
rpm -Uvh EMC-ScaleIO-sdc-x.x-x.x.rpm
```

NOTE: If the storage data client service fails after the PowerFlex gateway upgrade, reboot the storage data client node.

Upgrade hardware

When installing new nodes, ensure that you keep nodes with similar I/O characteristics in their own protection domain. Different CPUs, memory speeds, drive capacity, drive types, and network speeds can affect overall node performance.

PowerFlex Manager will not prevent mixing of node types within a protection domain, however, performance is dictated by the lowest performing system and that may not be desirable.

In the case of node replacement with newer nodes, add the new nodes to the existing protection domain, move your workloads, and MDM roles (if they exist) then remove the old nodes from the protection domain. Wait for the rebuild or re-balance processes to compete before adding or removing a node. For more information about adding and removing nodes or roles, see the *Dell PowerFlex and PowerFlex Manager Software Administration Guide*.

7

Post-upgrade configuration tasks

These configuration tasks are optional and should only be performed after upgrading a PowerFlex cluster.

The following tasks are explained in this section:

- Optionally configure the NVMe over TCP protocol.
- If you are using volume through NVMe over TCP protocol, configure the following manually after upgrading to PowerFlex Manager to use the complete load balancing feature:
  - Configure the system data networks of the system.
  - Configure the network sets.
  - Define host groups with their set of reachable networks.
  - Assign hosts to host groups.
  - Disconnect and connect NVMe and TCP subsystem with the persistent discovery option.

Related information

Upgrade workflow from PowerFlex 5.0.x and PowerFlex Manager 5.0.x and above to PowerFlex 5.1.x and PowerFlex Manager 5.1.x
Upgrade workflow from PowerFlex 4.5.x and PowerFlex Manager 4.8.x to PowerFlex 4.5.x and PowerFlex Manager 5.1.x

Configuring NVMe initiators as hosts connecting to PowerFlex system

Use the following procedures to configure the NVMe initiators on hosts for Linux and VMware ESXi-based systems.

The procedure may vary depending on the operating system versions. For more information, see the operating system vendor documentation.

NOTE: NVMe over TCP initiator host-side support is only available for VMware ESXi and SUSE. All other Linux platforms are considered as technology preview.

Configure NVMe initiators on hosts for Linux-based systems

For systems using NVMe over TCP technology, configure the NVMe initiator on all hosts that access PowerFlex storage. The following procedure is provided as an example only. Consult your operating system's documentation for detailed instructions about configuring NVMe initiators for NVMe over TCP topologies.

Prerequisites

Ensure that you have the login credentials for each host.

About this task

In an erasure coding environment, when NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node, and configure the NVMe targets in PowerFlex.

In a mirroring environment, when NVMe over TCP technology is used, PowerFlex software is not installed on the compute nodes. Instead, configure the NVMe initiators in the operating system on the PowerFlex compute-only node or PowerFlex hyperconverged node, and configure the NVMe targets in PowerFlex.

NOTE: Ensure that the hosts are running Linux operating system versions that are supported by this version of PowerFlex. For more information, see the system requirements in the *Dell PowerFlex and PowerFlex Manager Technical Overview*.

The following example shows one way of configuring the NVMe initiator. See your operating system's documentation for details and other configuration options.

Steps

1. Ensure that the NVMe modules are loaded on the NVMe initiator:
   a. Log in to the host.
   b. In the command line, run the command:

      ```
      lsmod | grep nvme
      ```

   c. Ensure that `nvme_tcp` and `nvme_fabrics` modules are loaded.
      If they are loaded, go to step 2. If they are not loaded, continue to the next substep.
   d. Perform one of the following procedures:
      - For manual loading of modules (perform after every reboot), run:

        ```
        modprobe nvme_tcp
        modprobe nvme_fabrics
        ```

      - For automatic loading of NVMe modules, run:

        ```
        echo nvme_tcp >> /etc/modules-load.d/nvme_tcp.conf
        echo nvme_fabrics >> /etc/modules-load.d/nvme_tcp.conf
        ```

2. Ensure that the NVMe CLI is installed on the initiator:
   a. Run the following command:

      ```
      nvme
      ```

      If the CLI responds, go to step 3. If it does not, go to the next substep.
   b. Run the command:

      ```
      dnf install nvme-cli
      ```

3. Ensure that a unique host NQN and unique host ID are configured in the initiator.
   a. To verify that a host NQN is present, run the command:

      ```
      nvme show-hostnqn
      ```

      Example of a host NQN:

      ```
      nqn.2014-08.org.nvmexpress:uuid:4c4c4544-0051-5610-8046-b5c04f484b32
      ```

   b. If no host NQN exists, generate one by running the command:

      ```
      nvme gen-hostnqn > /etc/nvme/hostnqn
      ```

      NOTE: Make a note of each PowerFlex node's host NQN. You need this information when configuring the NVMe targets with PowerFlex Manager.

   c. To verify that a host ID is present, run the command:

      ```
      cat /etc/nvme/hostid
      ```

      Example of a host ID:

      ```
      4c4c4544-0051-5610-8046-b5c04f484b32
      ```

      Last 36 characters of a nvme gen-hostqn name can be used. The format must be precise, otherwise the connection fails.
4. Perform the following actions for each required protection domain:
   a. Add an NVMe host to PowerFlex.

b. Create volume.
c. Map volume

For more information, see the *Dell PowerFlex and PowerFlex Manager Software Administration Guide* and *Dell PowerFlex and PowerFlex Manager CLI Reference Guide*.

5. Discover and connect to the target PowerFlex system.
6. Configure the type of multipath used:

Multipath type: Native NVMe Multipath

Use the following procedure:

- Check if the native NVMe multipath is enabled using the following command:

```
cat /sys/module/nvme_core/parameters/multipath
```

- If `Y` is returned, native NVMe multipath is already enabled.
- To enable the NVMe multipath, run the following command:

```
grubby --update-kernel=ALL --args="nvme_core.multipath=Y"
grub2-mkconfig -o /boot/grub2/grub.cfg
reboot
```

- To enable active/active protocol which remains persistent after reboots, run the following command:

```
echo 'ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{iopolicy}="round-robin"' >
/lib/udev/rules.d/71-nvme-iopolicy.rules
```

7. Configure a persistent initiator. This initiator automatically forms NVMe connections after reboots.
   a. Create a service that runs `nvme connect-all` on reboots:

```
vi /etc/systemd/system/nvme_fabrics_persistent.service
[Unit]
Description=NVMf auto discovery service
Requires=network.target
After=systemd-modules-load.service network.target
[Service]
Type=oneshot
ExecStart=/usr/sbin/nvme connect-all
StandardOutput=journal
[Install]
WantedBy=multi-user.target timers.target
```

   b. Start a new service, and enable it:

```
systemctl start nvme_fabrics_persistent.service
systemctl enable nvme_fabrics_persistent.service
```

8. Run the following command to update the udev rule to make the recovery delay 15 seconds:

```
echo ACTION!=\"remove\", KERNELS==\"ctl\", SUBSYSTEMS==\"nvme-fabrics\",
ATTR{transport}==\"tcp\", ATTR{model}==\"powerflex\", ATTR{recovery_delay}=\"15\" >>
/etc/udev/rules.d/71-nvme-recovery-delay.rules
```

NOTE: This step is required only for SUSE Linux Enterprise Server (SLES) 15.5 version.

Discover and connect to the target PowerFlex system

Use the following procedure to obtain SDT IP addresses for discovering the target PowerFlex system.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain SDT IP addresses.

Steps

1. From PowerFlex Manager, click **Block** > **Storage systems** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery port in the lower right corner.
4. To perform the discovery to check that there is an NVMe connection, use the following command:

   ```
   #nvme discover --transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
   trsvcid=8009
   ```

   192.168.163.17 is the NVMe over TCP controller IP address (data 1 VLAN) , and **192.168.163.251** is the NVMe over TCP host IP address (data 1 VLAN).

   ```
   Discovery Log Number of Records 4, Generation counter 2
   =====Discovery Log Entry 0======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  1
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.19
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 1======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  17
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.18
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 2======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  32
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.161.20
   eflags:  not specified
   sectype: none
   =====Discovery Log Entry 3======
   trtype:  tcp
   adrfam:  ipv4
   subtype: nvme subsystem
   treq:    not specified, sq flow control disable supported
   portid:  49
   trsvcid: 4420
   subnqn:  nqn.1988-11.com.dell:powerflex:00:c5ea88815e07e40f
   traddr:  192.168.163.17
   eflags:  not specified
   sectype: non
   ```

5. On the NVMe over TCP client, connect the client to the PowerFlex NVMe over TCP target system.
6. Configure the `/etc/nvme/discovery.conf file` to add the parameters used in the NVMe discover command and use `-p` at the end of the command for persistent controller discovery.

   Persistent discovery controller will:

   a. Notify a host of fabric changes after initial discovery.
   b. The host retains connection to discovery controller.

NOTE: Add an entry for all the SDT IP addresses discovered in the previous step:

```
echo "--transport=tcp --traddr=192.168.163.17 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.18 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.163.19 --host-traddr=192.168.163.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
echo "--transport=tcp --traddr=192.168.161.20 --host-traddr=192.168.161.251 --
trsvcid=8009 -p" >> /etc/nvme/discovery.conf
```

Where

- `--traddr=192.168.161.xx`— Data 1 IP address configured in SDT.
- `--traddr=192.168.163.xx`—Data 2 IP address configured in SDT.
- `--host-traddr=192.168.163.251`—Data 1 IP address configured in the NVMe host.
- `--host-traddr=192.168.161.251`—Data 2 IP address configured in the NVMe host.

7. Connect the NVMe over TCP host to the controller system.
   `# nvme connect-all`
8. Verify that the NVMe over TCP host can access the namespace:
   `# nvme list-subsys`

Configuring NVMe over TCP on a VMware ESXi compute-only node

When configuring NVMe over TCP on a VMware ESXi compute-only node, ensure the following:

- NVMe over TCP resource group is deployed.
- VMware ESXi compute-only nodes are deployed.
- Host is running VMware ESXi version 8.0 or higher.
- VMware vSphere distributed switch is at version 8.0 or higher.

Enable the NVMe over TCP VMkernel ports

Use this procedure to enable the NVMe over TCP VMkernel ports.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home/Inventory** and select the host.
3. Select **Configure** > **VMkernel adapters**.
4. Edit **PowerFlex-Data 1**.
5. Select the **NVMe over TCP** check box and click **OK**.
6. Repeat these steps for the remaining PowerFlex data networks and remaining hosts in the cluster.

Add NVMe over TCP software storage adapter

Use this procedure to add NVMe over TCP software storage adapter.

Steps

1. Log in to the VMware vSphere Client.
2. Click **Home** > **Inventory** > **Hosts and Clusters**.
3. In the VMware vSphere console, browse to the customer data center, compute-only cluster, and select the added host.
4. In the right pane, click **Configure** > **Storage Adapters**.
5. In the right pane, click **Add Software Adapter**.
6. Click **Add NVMe over TCP adapter**.
7. Select the first flex_dvswitch VMNIC and click **OK**.

NOTE: flex_dvswitch is the dvswitch created for flex-data1-<vlanid>, flex-data2-<vlanid> and flex-data3-<vlanid>, flex-data4-<vlanid>, if required.

8. Click **Add NVMe over TCP adapter**.
9. Select the second flex_dvswitch VMNIC and click **OK**.

NOTE: flex_dvswitch is the dvswitch created for flex-data1-<vlanid> and flex-data2-<vlanid>.

Copy the host NQN

Use this procedure to copy the host NQN to the copy buffer. The host NQN details are required when you add the host to the PowerFlex Manager.

Steps

1. Log in to VMware vSphere Client.
2. Select the first host.
3. From the right pane, select **Configure** > **Storage Adapters**.
4. Select the first VMware NVMe over TCP storage adapter. For example, **vmhba6x**.
5. From the pane, select **Controllers/Add Controller**.
   The host NQN is listed at the top of the form.
6. Click **Copy** and place the host NQN in the copy buffer.
7. Click **Cancel**.
8. Repeat the steps for all the hosts.

Add a host to PowerFlex

Use this procedure to add a host to PowerFlex using PowerFlex Manager.

Steps

1. Log in to PowerFlex Manager.
2. Click **Block** > **Hosts**.
3. Click **Add Host**.
4. Enter the hostname and paste the host NQN from the copy buffer.
5. Enter the **Number of Paths** per volume. The default number of paths are four per volume, and maximum number of paths are eight per volume.
6. Enter the **Number of System Ports** per protection domain. The default number of system ports are 10 per protection domain, and maximum number of system ports are 16 per protection domain.
7. Click **Add**.

Create a volume

Use this procedure to create a volume.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Click **Create Volume**.
3. Enter the number of volumes and the name of the volumes.
4. Select **Thick** or **Thin**. Thin is the default.
5. Enter the required volume size in GB, in 8 GB increments.
6. Select the storage pool and click **Create**.

Map a volume to the host

Use this procedure to map a volume to the host.

Steps

1. From PowerFlex Manager, click **Block** > **Volumes**.
2. Select the **Volume** check box and click **Mapping** > **Map**.
3. Select the protocol NVMe.
4. Select the check box on the host that you are mapping the volume to.
5. Click **Map**.

Discover and connect the NVMe over TCP target

Use this following procedure to discover and connect the NVMe over TCP target PowerFlex system using VMware ESXi.

Prerequisites

- Ensure that a volume is mapped to the host to be able to connect the SDT paths.
- Obtain the SDT IP addresses.
- Use the esxcli command to perform the procedure.

Steps

1. From PowerFlex Manager, click **Block** > **Storage System** > **NVMe Targets**.
2. Select any one of the listed SDTs.
3. Record the IP addresses and discovery ports in the lower right corner.
   Example output:

   Table 9. Record IP addresses and discovery ports

IP address	Storage port	I/O port	Discovery port
192.168.151.2	12200	4420	8009
192.168.152.2	12200	4420	8009
192.168.153.2	12200	4420	8009
192.168.154.2	12200	4420	8009

4. SSH to the VMware ESXi server.
5. Run a discovery query on each adapter enter:
   ```
   #esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009
   #esxcli nvme fabrics discover -a vmhba6y -i 192.168.y.y -p 8009
   ```
   - Where **6x** is the first NVMe over TCP software adapter, and **6y** is the second NVMe over TCP software adapter.
   - Where **192.168.x.x** is first IP address, and **192.168.y.y** is second IP address of any one of the SDTs recorded.

   NOTE: The IP addresses used are for EPS form, this is an example only, see the EPS form for the IP addresses for your PowerFlex system.

6. Connect to PowerFlex system:
   ```
   #esxcli nvme fabrics discover -a vmhba6x -i 192.168.x.x -p 8009 -c
   #esxcli nvme fabrics discover -a vmhba6y -i 192.168.x.y -p 8009 -c
   ```
7. Verify the connected controllers enter:
   ```
   #esxcli nvme controller list
   ```

Perform a rescan of the storage

Use this procedure to perform a rescan of the storage.

Steps

1. Log in to VMware vSphere Client.
2. Browse to a host, a cluster, a data center, or a folder that contains hosts.
3. Right-click the menu, select **Storage** > **Rescan Storage**.
4. Select **Scan for new Storage Devices** and click **OK**.

Create a new VMFS datastore

Use this procedure to create a new virtual machine file system (VMFS) datastore.

Steps

1. Log in to the VMware vCenter.
2. Click **Home** > **Inventory** and select the storage icon.
3. Right-click the compute-only cluster and select **Storage** > **New Datastore**.
4. Click **VMFS** and click **Next**.
5. Enter the name of the datastore and select a host from the list.
   Verify if you see a new NVMe TCP disk (eui.######).
6. Select the NVMe TCP disk and click **Next**.
7. Click **VMFS 6** and click **Next**.
8. Leave the default configuration unless instructed by the customer and click **Next**.
9. Review the configuration and click **Finish**.

Configuring system data networks, network sets, hosts, host groups, and enabling persistent discovery (Optional)

After upgrading, optionally configure the following to enable the load-balancing feature on a volume that is using the NVMe over TCP protocol:

- System data networks
- Network sets
- Host groups with their set of reachable networks
- Assign hosts to host groups

Persistent discovery with NVMe over TCP

To enable persistent discovery controller option on a Linux-based compute-only node that is mapped to an NVMe over TCP volume, run the following:

Table 10. Persistent discovery with NVMe over TCP

To...	Do this...
Discover the PowerFlex NVMe storage data target interfaces.	For a root user: **`#sudo nvme discover -t tcp -a <SDT IP ADDRESS> -s 4420`** For a non-root user: **`#sudo nvme discover -t tcp -a <SDT IP ADDRESS> -s 4420`**
Connect all storage data target paths. Run the same command for all the discovered storage data target IP addresses (**`traddr`**).	For a root user: **`#nvme connect-all -t tcp -a <SDT IP ADDRESS> -p`**

Table 10. Persistent discovery with NVMe over TCP (continued)

To...	Do this...
	For a non-root user: `#sudo nvme connect-all -t tcp -a <SDT IP ADDRESS> -p`

Define system data networks

When the resource group is deployed, configure system data networks/subnets to be used to connect hosts initiator to the storage data target (SDT).

Prerequisites

- Maximum allowed system data networks are eight.
- Configure two or four system data networks for PowerFlex depends on the number of PowerFlex data (SDS-SDS and SDS-SDT communication for a mirroring data protection method) or (storage node-storage node and storage node-SDT communication for an erasure coding data protection method) networks configured for the deployment.
- If you have not defined the system data network, by default, a host can reach all system data networks. Ignoring the data networks/subnets may result in unequal load between the host initiator ports and therefore non-optimized IO performance with the PowerFlex system. In addition, it may impact the path resiliency if not all the host initiator ports are able to connect to the system.
- Configure the routing tables, if the system is a Layer-3 network.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **Networks**.
3. From the **Networks** page, click **Define** > **Network Type** and select **PowerFlex Data**.
4. Complete the required information and click **Save**.

Define host group and assign hosts

Use this procedure to define a host group and assign hosts to the host group using SCLI. This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

About this task

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. In this release, host group has only limited functionality, as required for the NVMe over TCP feature. The maximum number of host groups allowed per system is 1024.

By default, a host is always part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups.

Steps

1. Define the host groups with the associated reachable networks, type:

   ```
   scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]
   ```

2. Assign hosts to the host group, type:

   ```
   scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)
   ```

Define network sets

Use this procedure to define the network sets using SCLI. This feature is allowed only via SCLI. It is not available in PowerFlex Manager.

About this task

A set of networks that are connected to the same switch, which is configured for balancing the I/O traffic between access switches by associating system data networks to network set. Load balancer will take care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2. Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network is considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network set is applicable for a PowerFlex deployment.

Steps

Configure the network set using CLI, type:

```
scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]
```

Configuring system data networks, network sets, hosts, host groups, and enabling persistent discovery (Optional)

After upgrading, optionally configure the following to enable the load-balancing feature on a volume that is using the NVMe over TCP protocol:

- System data networks
- Network sets
- Host groups with their set of reachable networks
- Assign hosts to host groups

Persistent discovery with NVMe over TCP

To enable persistent discovery controller option on a Linux-based compute-only node that is mapped to an NVMe over TCP volume, run the following:

Table 11. Persistent discovery with NVMe over TCP

To...	Do this...
Discover the PowerFlex NVMe storage data target interfaces.	For a root user: **`#sudo nvme discover -t tcp -a <SDT IP ADDRESS> -s 4420`** For a non-root user: **`#sudo nvme discover -t tcp -a <SDT IP ADDRESS> -s 4420`**
Connect all storage data target paths. Run the same command for all the discovered storage data target IP addresses (**`traddr`**).	For a root user: **`#nvme connect-all -t tcp -a <SDT IP ADDRESS> -p`** For a non-root user: **`#sudo nvme connect-all -t tcp -a <SDT IP ADDRESS> -p`**

Define system data networks

When the resource group is deployed, configure system data networks/subnets to be used to connect hosts initiator to the storage data target (SDT).

Prerequisites

- Maximum allowed system data networks are eight.
- Configure two or four system data networks for PowerFlex depends on the number of PowerFlex data (SDS-SDS and SDS-SDT communication for a mirroring data protection method) or (storage node-storage node and storage node-SDT communication for an erasure coding data protection method) networks configured for the deployment.
- If you have not defined the system data network, by default, a host can reach all system data networks. Ignoring the data networks/subnets may result in unequal load between the host initiator ports and therefore non-optimized IO performance with the PowerFlex system. In addition, it may impact the path resiliency if not all the host initiator ports are able to connect to the system.
- Configure the routing tables, if the system is a Layer-3 network.

Steps

1. Log in to PowerFlex Manager.
2. Click **Settings** > **Networks**.
3. From the **Networks** page, click **Define** > **Network Type** and select **PowerFlex Data**.
4. Complete the required information and click **Save**.

Define host group and assign hosts

Use this procedure to define a host group and assign hosts to the host group using SCLI. This feature is allowed only using SCLI. It is not available in PowerFlex Manager.

About this task

The host group holds a subgroup of existing networks that are accessible to a group of hosts. Each host group is associated with a set of system networks. Each host may be associated with multiple host groups. In this release, host group has only limited functionality, as required for the NVMe over TCP feature. The maximum number of host groups allowed per system is 1024.

By default, a host is always part of the default host group until you create a new host group and associate the host to the new host group. By default, a host supports all data networks and you do not need to configure the host groups.

Steps

1. Define the host groups with the associated reachable networks, type:

   ```
   scli --add_host_group [--host_group_name <NAME>] [(--network_id <ID> | --network_name <NAME>)] [(--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>)]
   ```

2. Assign hosts to the host group, type:

   ```
   scli --assign_host_to_host_group (--host_nqn <NQN> | --host_id <ID> | --host_name <NAME>) (--host_group_id <ID> | --host_group_name <NAME>)
   ```

Define network sets

Use this procedure to define the network sets using SCLI. This feature is allowed only via SCLI. It is not available in PowerFlex Manager.

About this task

A set of networks that are connected to the same switch, which is configured for balancing the I/O traffic between access switches by associating system data networks to network set. Load balancer will take care of the distribution of networks over network sets.

For example, if a system is defined with two system data networks (192.168.150.0 and 192.168.151.0), then you could create two network sets and associate 192.168.150.0 with network set 1 and 192.168.151.0 to network set 2. Each system data network belongs to exactly one network set. If the network was not assigned to a network set, the network is considered a network set of its own.

Use the default network set and do not manually create the network set. Creating network set is applicable for a PowerFlex deployment.

Steps

Configure the network set using CLI, type:

```
scli --add_network_set [--network_set_name <NAME>] [(--network_id <ID> | --network_name <NAME>)]
```